



● Ahmad Ibrahim · Product design portfolio

Open to new roles · 5+ years · Cape Town, South Africa (UTC+2) · Canadian citizen · remote

I design data-heavy product interfaces, and build them.

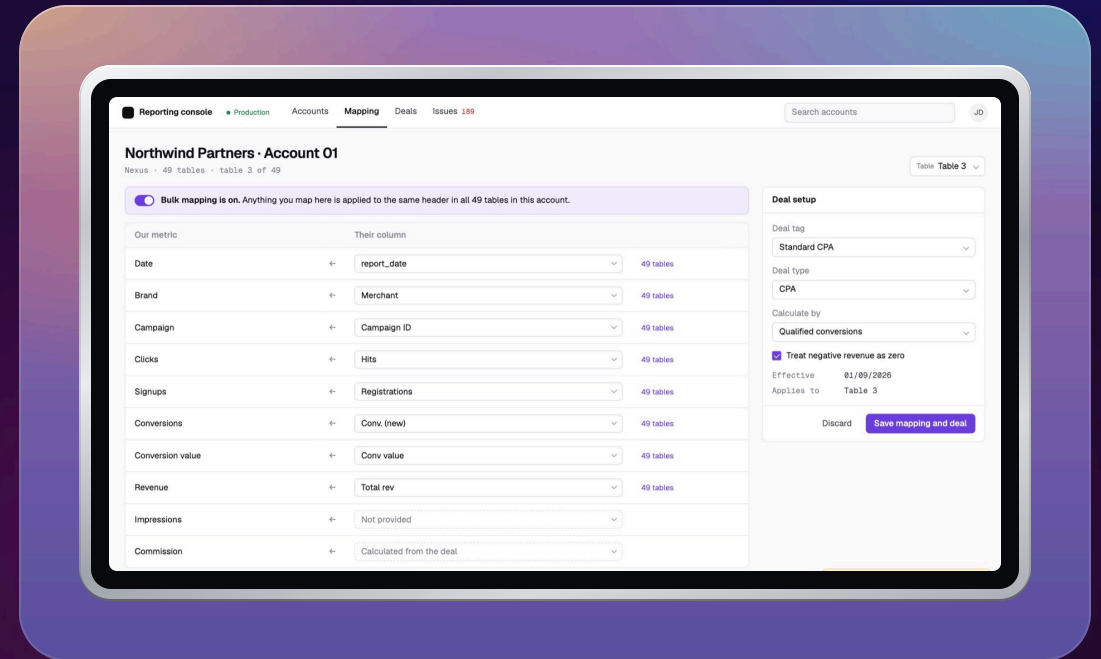
I design dashboards and internal tools for technical teams, then build them as working front ends.

[View work](#)

[ahmad-ibrahim.com ↗](https://ahmad-ibrahim.com)

[Email](#)

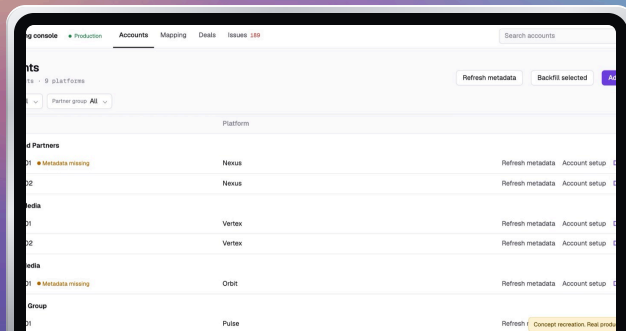
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Featured Internal reporting tools
Concept recreation, real product under NDA

[Contents](#)

Selected work



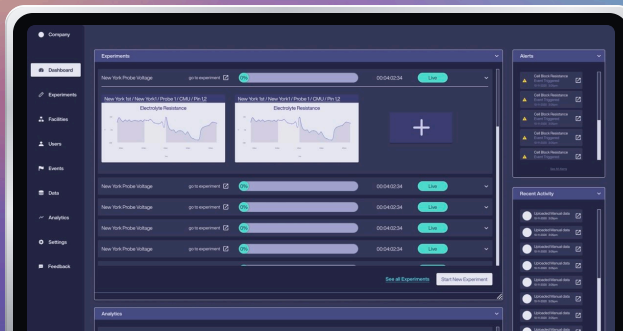
In use

Page 5

Internal reporting tools

Canadian marketing-technology company

Tools that replaced manual reporting. The team no longer logs in to every source and copies the data into spreadsheets, which saves each person an estimated 10–20 hours a week. Happy to walk you through it in an interview.

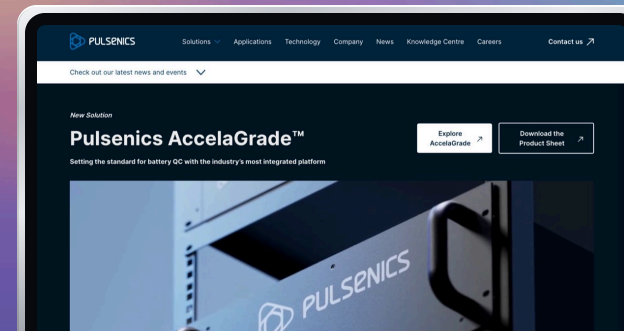


In use

Page 12

Pulsenics System Software

The app Pulsenics' clients use to manage their probes, run experiments and analyse the data. It's part of the company's core offering.



Live

Page 21

Pulsenics Website

With Marc Varone, ran a workshop with the marketing team that led to an Applications page, where visitors work out which product they need. I built the Knowledge Centre for publishing what Pulsenics knows about EIS, within the site's original visual style.

● About me

Ahmad Ibrahim

I started in a data role at a Canadian marketing-technology company. I learned everything I could about data and database management, ended up owning part of the data funnel, and then used my UX skills to make that work easier. Today I lead a small reporting team and design the internal tools it and the company's other reporting teams use. My background is industrial design (OCAD University).

The thing I care about most in an interface: when you land on it, it should be clear what it's for and how to use it.

✕ Skills

Focus	Data-heavy B2B products and internal tools
Design	Figma, Figma Make, design systems
Build	React, HTML/CSS, Webflow, Claude Code, Codex
Data	SQL and reporting
Hours	Cape Town (UTC+2). Available until 10pm, so I overlap with US and Canada Eastern until about 3pm their time

✕ Earlier work

Knowledge Leap

My OCAD University thesis: an open source AR app for Magic Leap that turns your surroundings into lessons.

[View ↗](#)**DuoVillage**

A group research project: a Duolingo concept for practising real conversations without the social anxiety.

[View ↗](#)



● How I work

1 Understand the problem

I talk to the people who'll use it, watch how they work now, and run workshops or card sorts with the team.

See: [card sort with Pulsenics staff](#), [marketing workshop](#)

2 Map the flow

From the first step to the last, before any screens.

See: [the experiment flow](#)

3 Sketch, test and refine

Rough sketches become rough mockups. When time and resources allow, I put them in front of users, then refine over several rounds until the interface is clear and works.

See: [wireframes V1 to V8](#), [testing the redesign with my team](#)

4 Design system, then build

I set up the design system and foundations in Figma. That makes the full interface faster to build, and it gives AI a system to work from. From there I use AI to build working interfaces for another round of testing, then the full app, and I keep refining.

See: [the Pulsenics UI system](#), [the reporting redesign](#)

× Where AI fits

I run the research myself. It's conversations, workshops and watching people work, and I want to be part of all of it. AI helps me organise what I find and spot patterns I might have missed.

Once the structure and flows are solid, AI works from my Figma design system. I use it for the detail work that takes hours by hand: tuning corner radii, refining the colour palette, adjusting spacing. It's also how I explore. I can try variations or change the visual direction, and the whole interface follows without starting over.

Case study 01

Status
In use, under NDA

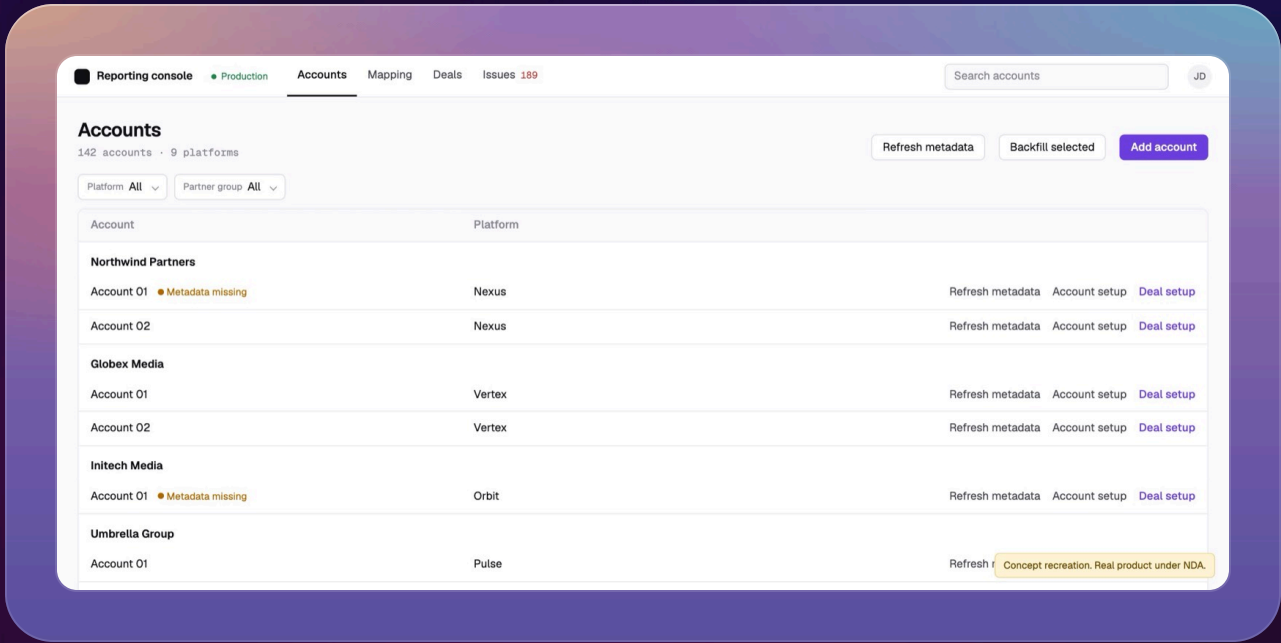
Work
Product design

Product
Internal data tools

Internal reporting tools

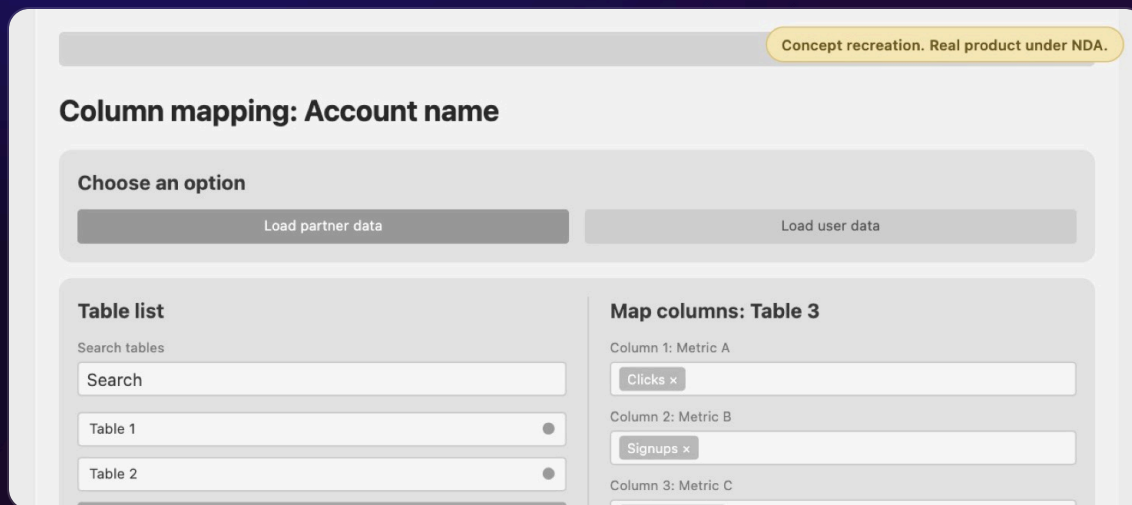
A console that collects performance data from hundreds of partner accounts across dozens of platforms, so the reporting team no longer copies numbers into spreadsheets by hand.

Client	Canadian marketing-technology company (under NDA)
Product	Internal tools for collecting data and building reports
My role	Sole designer. UI and design system, first builds in Figma Make, data flows, validation and testing
Scope	Lead a team of three and support a second team. The tools serve about 12 people; I scope them with the PMO, two developers, a DBA and infrastructure.
Timeline	First version shipped late 2023. Redesign shipped in 2025, in about two months



The account directory. Concept recreation. Real product under NDA.

An estimated 10–20 hours a week back for each person on the reporting team.



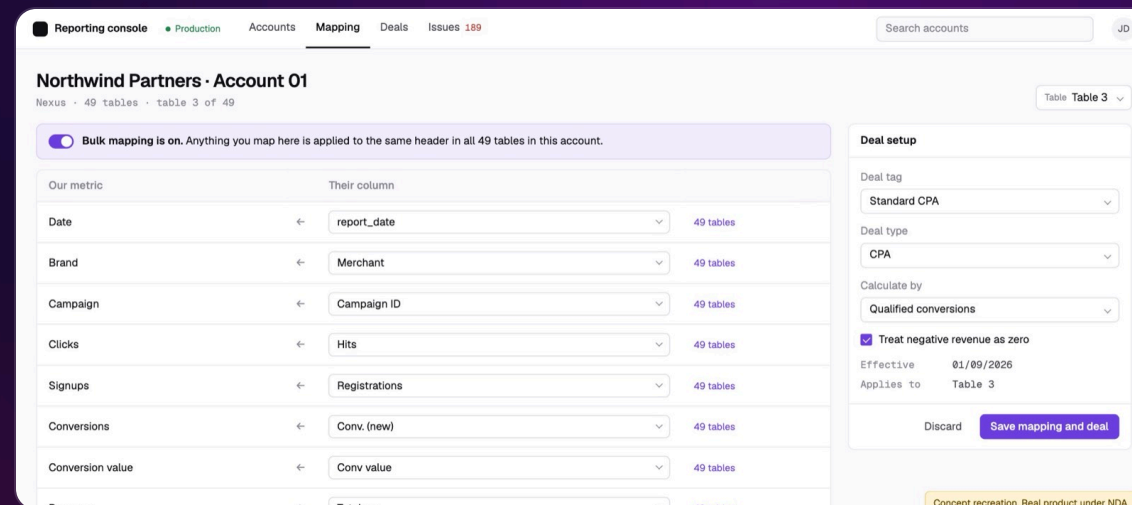
Version one, 2023. Every table mapped by hand.

Both screens are concept recreations with fake data. The real product is under NDA.

● The problem

In plain terms: the company works with hundreds of partners. Each one gives us access to an account on their own reporting platform, where we can see numbers like clicks, signups and revenue. Every platform names and lays out those numbers differently, and each partner has a deal that sets how the numbers turn into what they pay.

We filled out performance reports by logging in to every account and entering the data into a spreadsheet by hand. That data was combined and sent as reports to managers, commercial marketing leads and executives, who used it to make finance decisions.



The redesign, 2025. Map one table and the whole account follows.

● How we worked

I spent a week on site with the developers for an intensive build. I designed the UI and explained how the data flows. I also researched the API documentation for the platforms that provide it and tested the data for accuracy. My team and I have kept that process going as the system has grown.

Version one, late 2023

It had to be built in Streamlit, so I could only use its components. The bigger problem was the workflow: every element of every table had to be mapped by hand.

The redesign, 2025

I redesigned the UI and its design system in Figma, then built the first working versions myself in Figma Make. I walked it through the functionality each screen needed, step by step, checked every element it produced and had it go back and fix what wasn't right. I tested the user flow with my team and kept refining with their feedback until we were happy with how it worked. Then we moved the build to Claude Code, where the developers took it to production. It shipped in about two months. The first version took months and was never as good as we needed.

Concept recreation. Real product under NDA.

Column mapping: Account name

Choose an option

Load partner data

Load user data

Table list

Search tables

Search

Table 1

Table 2

Table 3

Table 4

Table 5

Table 6

Table 7

Table 8

Table 9

Table 10

Table 11

Table 12

Map columns: Table 3

Column 1: Metric A

Clicks x

Column 2: Metric B

Signups x

Column 3: Metric C

Conversions x

Column 4: Metric D

None x

Column 5: Metric E

None x

Column 6: Metric F

Qualified x

Column 7: Metric G

None x

Column 8: Metric H

Revenue x

Deal setup

Table

Table 3

Deal tag

Deal tag

Deal type

Select deal type

Back

Refresh metadata

Save mapping

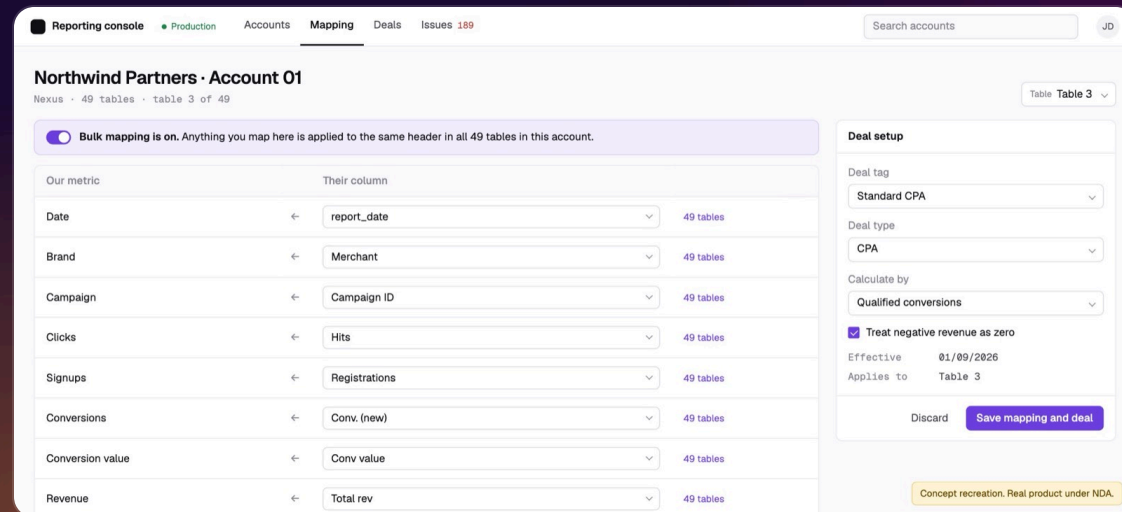
Version one, redrawn in greybox with generic labels.
Concept recreation. Real product under NDA.

Design decisions

01

Map one table, and the whole account is mapped

Most platforms use fixed headers that a model recognises automatically. For those that let owners write their own, the user maps a header's wording once and the tool applies it to every table that uses it. Copying mappings between accounts doesn't work, because no two are set up alike.

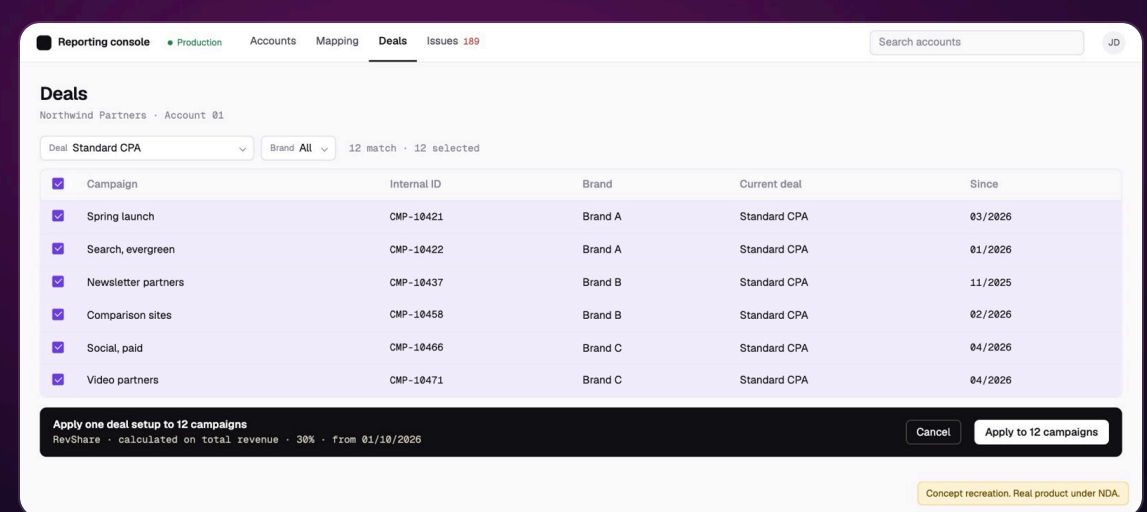


Bulk mapping on. Concept recreation. Real product under NDA.

02

One deal setup for many campaigns

Search campaigns by their deal and apply one setup to all of them: to ticked campaigns or tables, to everything a search returns, or, with an empty search, to the whole account.

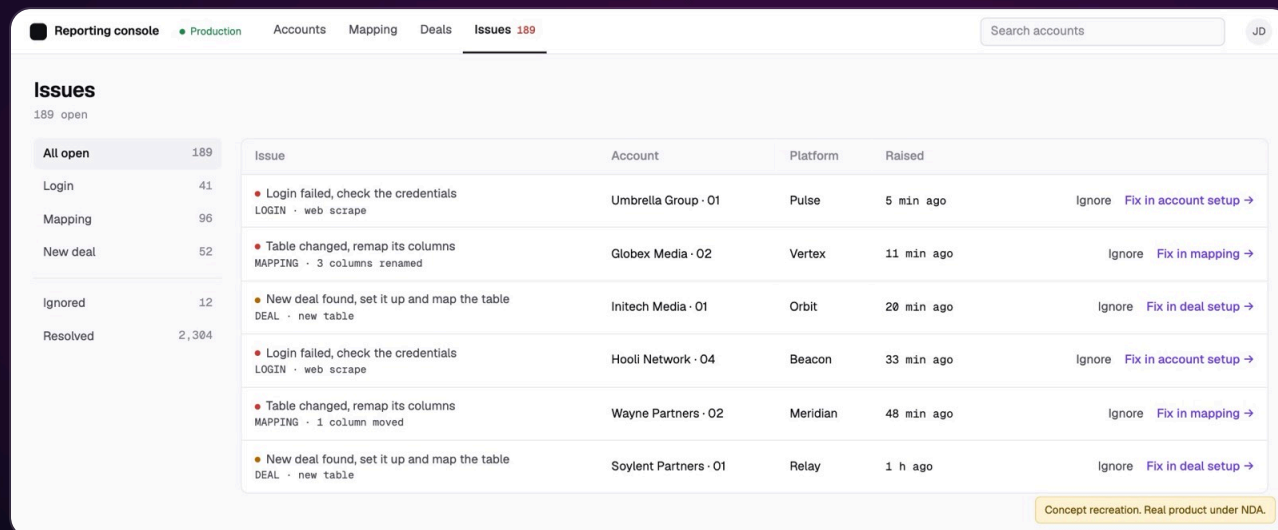


Bulk deal edit. Concept recreation. Real product under NDA.

03

A notification centre that keeps data collection running

It flags anything that would stop data coming in: login errors, such as wrong credentials on a platform we read through web scraping instead of an API; a table that changed and needs its columns remapped; a new deal that needs setting up and mapping. Each notification takes you straight to the account setup or mapping screen where the fix happens, and clears itself once it's fixed. Before, a Slack bot posted issues and someone had to react to mark them done.



The screenshot shows a web application interface for a 'Reporting console'. The top navigation bar includes 'Production', 'Accounts', 'Mapping', 'Deals', and 'Issues 189'. The 'Issues' section is active, showing a list of 189 open issues. The issues are categorized by status: All open (189), Login (41), Mapping (96), New deal (52), Ignored (12), and Resolved (2,384). The main table lists specific issues with columns for Issue, Account, Platform, and Raised. Each issue entry includes a description, the account name, the platform, the time raised, and actions like 'Ignore' and 'Fix in account setup'.

Issue	Account	Platform	Raised
• Login failed, check the credentials LOGIN · web scrape	Umbrella Group · 01	Pulse	5 min ago
• Table changed, remap its columns MAPPING · 3 columns renamed	Globex Media · 02	Vertex	11 min ago
• New deal found, set it up and map the table DEAL · new table	Initech Media · 01	Orbit	20 min ago
• Login failed, check the credentials LOGIN · web scrape	Hooli Network · 04	Beacon	33 min ago
• Table changed, remap its columns MAPPING · 1 column moved	Wayne Partners · 02	Meridian	48 min ago
• New deal found, set it up and map the table DEAL · new table	Soylent Partners · 01	Relay	1 h ago

The notification centre. Concept recreation. Real product under NDA.

04

Built for platforms that don't agree

Some platforms let account owners rename and move any column, so the only reliable way to match their columns to our metrics is for a person to assign them.

Some give total revenue but no commission column, so you have to calculate it yourself. Deals can vary within the same platform, and sales sometimes structure a deal differently from how it appears in the platform. That's why deal setup has to be fully flexible.

Some platforms limit how many calls you can make, so the daily loader pulls data at a custom rate for each one. We tried fixing things at the source, but the accounts belong to our partners, so the tools had to handle the variety.

Three platforms, one set of metrics

Platform A → *Columns renamed and moved* →

A person maps it

Platform B → *No commission column* →

Deal setup calculates it

Platform C → *Call limits* → Loader pulls at its own rate

Illustration with generic platform names.

05

Check it against the old reports

We ran the new reports alongside the old ones on historical data and released the data once accuracy passed an agreed threshold. My team and I signed off. The team still rotates through accounts checking the data, and each round comes back more accurate.

Metric	Spreadsheet	Console	
Clicks	48,210	48,210	Match
Signups	3,904	3,904	Match
Revenue	61,775.40	61,775.40	Match
Commission	18,532.62	18,540.12	Check the deal setup

Illustration with made-up numbers. A mismatch sends you back to the mapping or the deal before the switch.

Outcome

Time

An estimated 10–20 hours a week saved

For each person on the reporting team, because nobody logs in to sources and pastes data into spreadsheets any more.

Quality

Fewer errors and manual fixes, and faster reports

We know from our QA. Human error used to be common; now the automation collects the data and a person who understands it checks it. It also takes fewer people to run.

Upkeep

Bugs get caught and fixed faster

Thanks to the notification centre and the shorter build cycle. Most changes now are small improvements the team asks for.

Team

My team moved from data entry to QA

As the tools took over collecting the data, I moved my team from manual data entry to quality assurance. I got them SQL training so they could query the database and check the data themselves.



✕ Reflection

Looking back

Documenting as we went paid off

We kept notes on every platform the whole way: logins that fail on platforms we read by web scraping, owners who rename their columns, deals that sales set up differently from how the platform shows them. That record is what let us redesign quickly instead of relearning each platform.

Map the whole flow before the screens

Version one's real problem wasn't the screens. It was that every table had to be set up by hand. Mapping the full setup flow earlier, with the developers, would have brought bulk mapping and bulk deal edits into the first version.

Keep the build loop short

The first version took months. The redesign shipped in about two months, and problems now get fixed faster. Most of what we add now are small improvements the team asks for.

[Read it on the site ↗](#)

[Next: Pulsenics System Software](#)

Case study 02

Status
In use

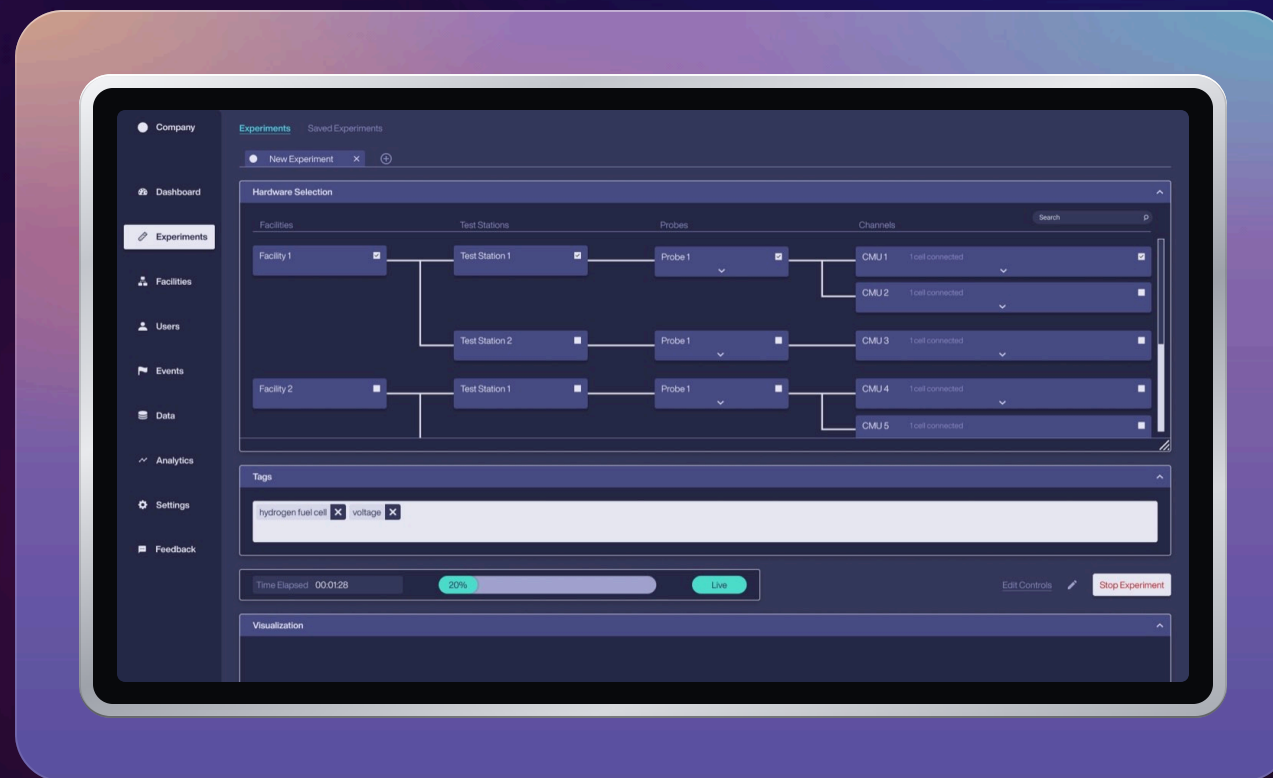
Work
Product design, design system

Product
Web app

Pulsenics System Software

The web app Pulsenics and its clients use to manage their testing hardware, run experiments, control probes and analyse electrochemical data.

Client	Pulsenics, Toronto
Product	Web app for running experiments and managing probes
Team	Me and Marc Varone (50/50), with Pulsenics' backend team
My role	Product design and the UI system, designed with Marc Varone from the ground up
Timeline	Designed from early 2021, shipped a year later
Status	In use



The Experiments view

Overview

Before

Code sent straight to the probes

The team ran experiments from testbed software, sending commands to the hardware in code.

What we did

Made those steps visual

We met the engineers, mapped how their experiments need to run, and designed each part of the app and its UI system.

Result

Shipped with the product launch

Pulsenics' engineers and their clients use it today across battery, hydrogen and fuel cell work.

The brief

Design a tool that lets Pulsenics and its clients run experiments and manage every probe they have connected.

Pulsenics makes electrochemical impedance spectroscopy (EIS) hardware: probes that test the health and performance of batteries, fuel cells and electrolyzers without damaging them. The software sits between that hardware and the people using it. It has to know what is plugged in where, start and stop measurements, and turn a stream of readings into something an engineer can act on.

Users

Technical teams

Engineers and researchers who know the chemistry and want precise control.

Data

Dense and continuous

Measurements stream in from many cells at once, across frequency and time, for hours or days.

Hardware

Physically nested

Facilities hold test stations, test stations hold probes, and probes connect to cells through channels.

How we got there

From commands in code to a visual UI

The app launched with the product. Until then, running an experiment meant using testbed software and sending commands to the probes in code. Our goal was to take that series of steps and make it visual.

1 Meet the engineers

We sat down with them and saw how they need an experiment to run.

2 Map the flow

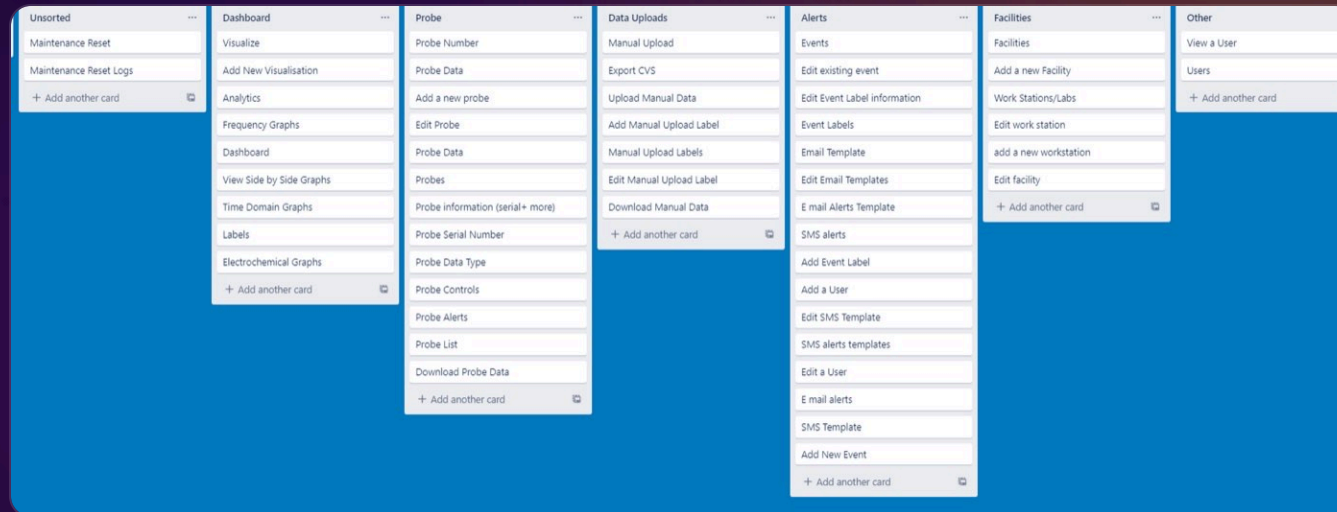
We mapped it from plugging in hardware to analysing the results.

3 Build it section by section

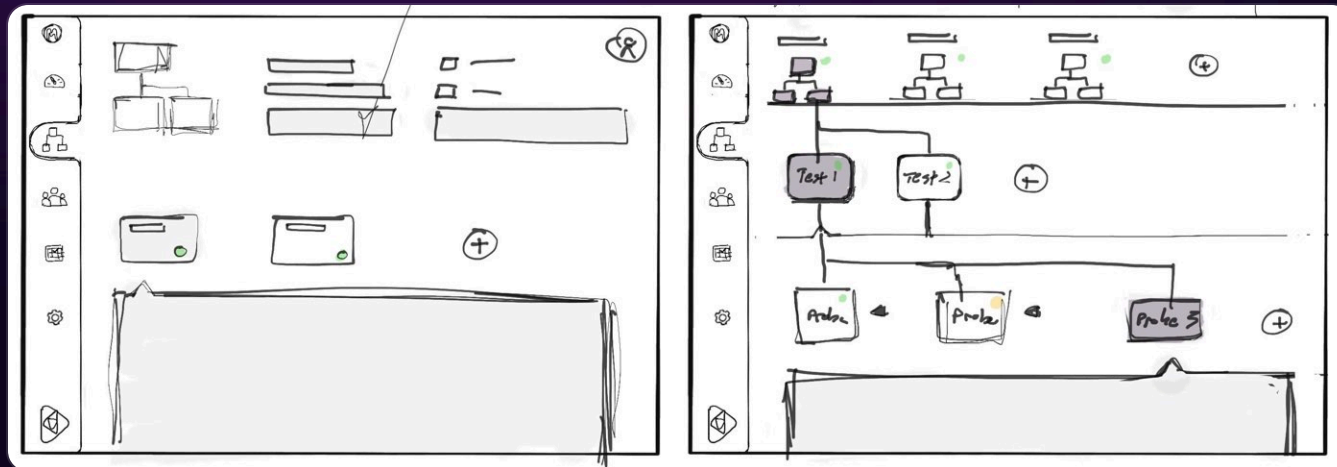
Then we designed each part in turn, from wireframes to final UI.

We worked closely with the backend team the whole way, so we understood quickly what was and wasn't possible. There was pushback on both sides at times.

The Figma file still shows the path: an open card sort with Pulsenics staff and user flows, sketches, eight rounds of wireframes, a foundation for the UI system, then the final screens and handoff to developers.

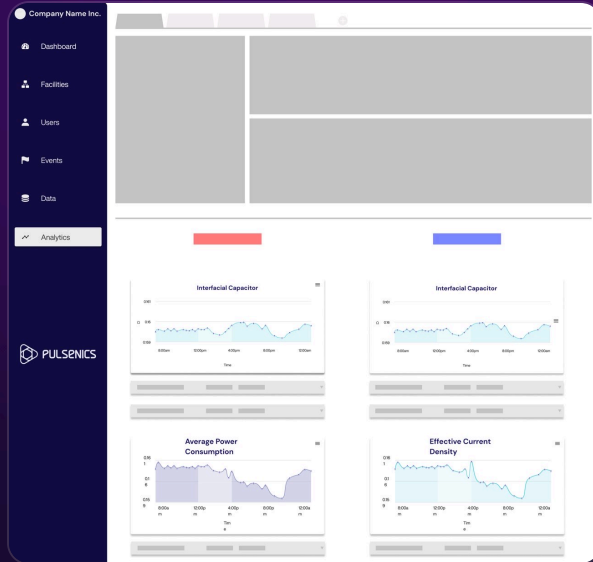


An open card sort with Pulsenics staff, from the CEO and engineers to the marketing team

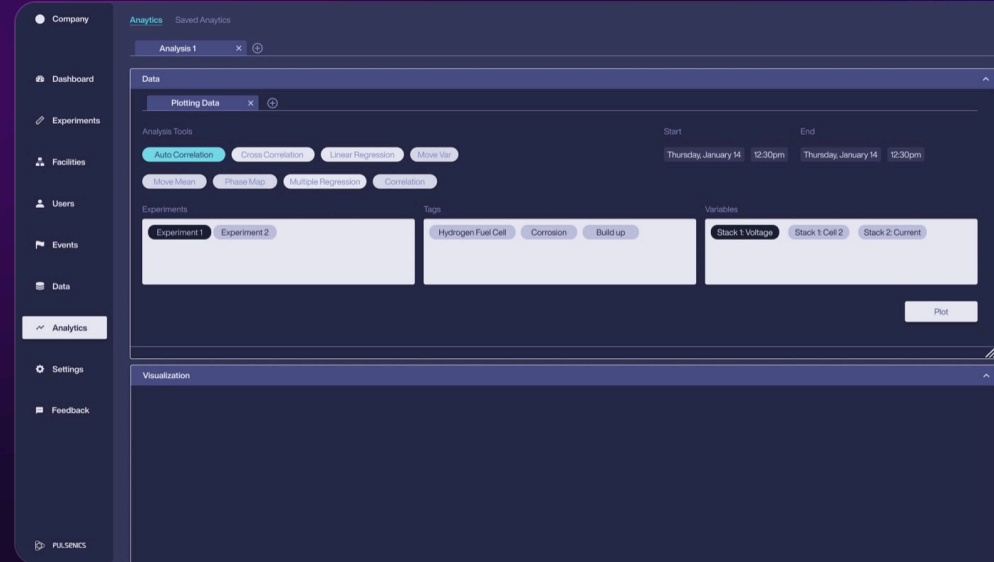


Sketches by [Marc Varone](#). The hardware tree on the right became decision 01.

Wireframes, V1 to V8



Wireframe V1: analytics as a page of fixed charts



Wireframe V8: analysis built from experiments, tags and variables. The final version is decision 05.

One flow, end to end

This is the flow we mapped with the engineers. Everything in the app supports it, from plugging in hardware to explaining what the data shows.

- 01 Pick the hardware
- 02 Start the experiment
- 03 Control the probe
- 04 Watch it live
- 05 Mark what happened
- 06 Analyse the results

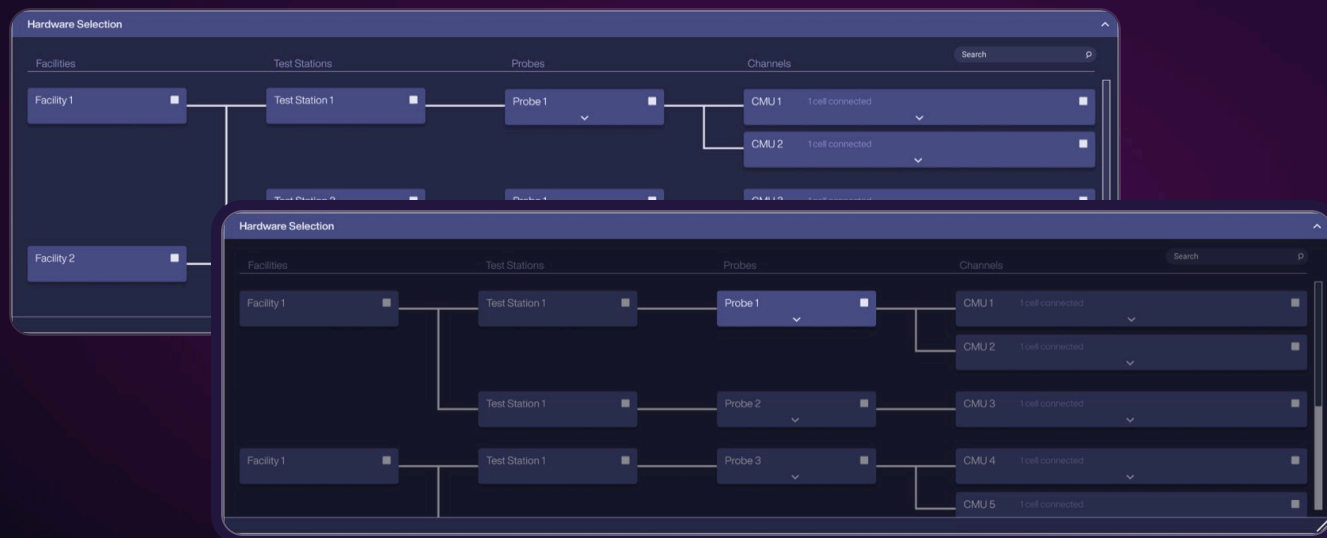
Key design decisions

01

Mirror the physical setup

There was no interface before this; experiments were set up in code. Hardware is chosen from a connected tree that matches how the equipment is actually wired: Facilities, then Test Stations, then Probes, then Channels, with each channel showing how many cells are connected.

Selecting any node dims every branch it doesn't belong to, so it's always clear which cells an experiment will touch. Search sits on top for setups too large to scan. Ticking a probe ticks its whole path, and each channel opens into the cells connected to it, which you can name, so you still know where everything is across several probes. Every facility, station, probe and channel can be renamed in place, and each probe shows its serial number.



The full tree, and the focus state when a probe is selected

02

Make starting an experiment one decision

Once hardware is picked, starting is a small pop-up: a start and end time, with two shortcuts for the most common cases, *Start Now* and *No specified end time*. It can stay that small because the real setup happens before it: choosing hardware and cells, adding tags and setting the controls. By the time you start, the only question left is when.

Start Experiment

Start

Thursday, January 14 12:30pm

End

Friday, January 14 12:30pm

☐ Start Now ☐ No specified end time

Start Experiment

Start Experiment

03

Give direct control of the instrument

Probe controls expose what the hardware actually does: choose a probe, set the time between samples, switch between galvanostatic and potentiostatic mode, and enter up to ten frequency and amplitude pairs. The engineers wanted the frequency and amplitude pairs entered in order, so they're a numbered list of fields rather than separate settings screens.

Send Controls and a red *Stop Sampling* sit together at the end, so the safe way out is always visible. Tabs keep the controls history, alerts and maintenance logs one click away. The maintenance log records who reset a probe, when and why, so a lab can trace any change to its equipment.

Number	Frequency (Hz)	Amplitude (A)	Reset
1	800	0.1	Reset
2			Reset
3			Reset
4			Reset
5			Reset
6			Reset
7			Reset
8			Reset
9			Reset
10			Reset

Probe controls

04

Keep running work in view

Live experiments get their own table with facility, test station, probe, timing and a progress bar, plus quick actions to open, duplicate or delete. Once a run finishes, it moves to Saved Experiments, so the live table only shows what's actually running. Running and finished experiments are kept apart on purpose.

Experiment	Facility	Test Station	Probes	Start Date	End Date	Status	
New Experiment	Facility	Test Station	Probe	12/06/2020 7:41 AM	12/06/2020 7:41 AM	0%	🔍 📄 🗑️
New Experiment 2	Facility	Test Station	Probe	12/06/2020 7:41 AM	12/06/2020 7:41 AM	0%	🔍 📄 🗑️
New Experiment 3	Facility	Test Station	Probe	12/06/2020 7:41 AM	12/06/2020 7:41 AM	0%	🔍 📄 🗑️
New Experiment 4	Facility	Test Station	Probe	12/06/2020 7:41 AM	12/06/2020 7:41 AM	0%	🔍 📄 🗑️

Live experiments

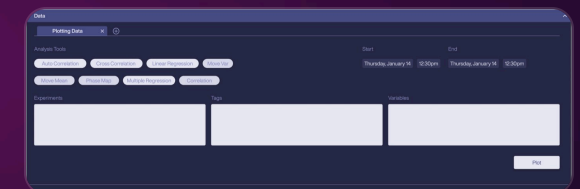
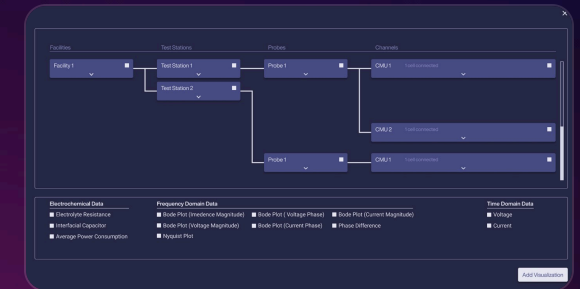
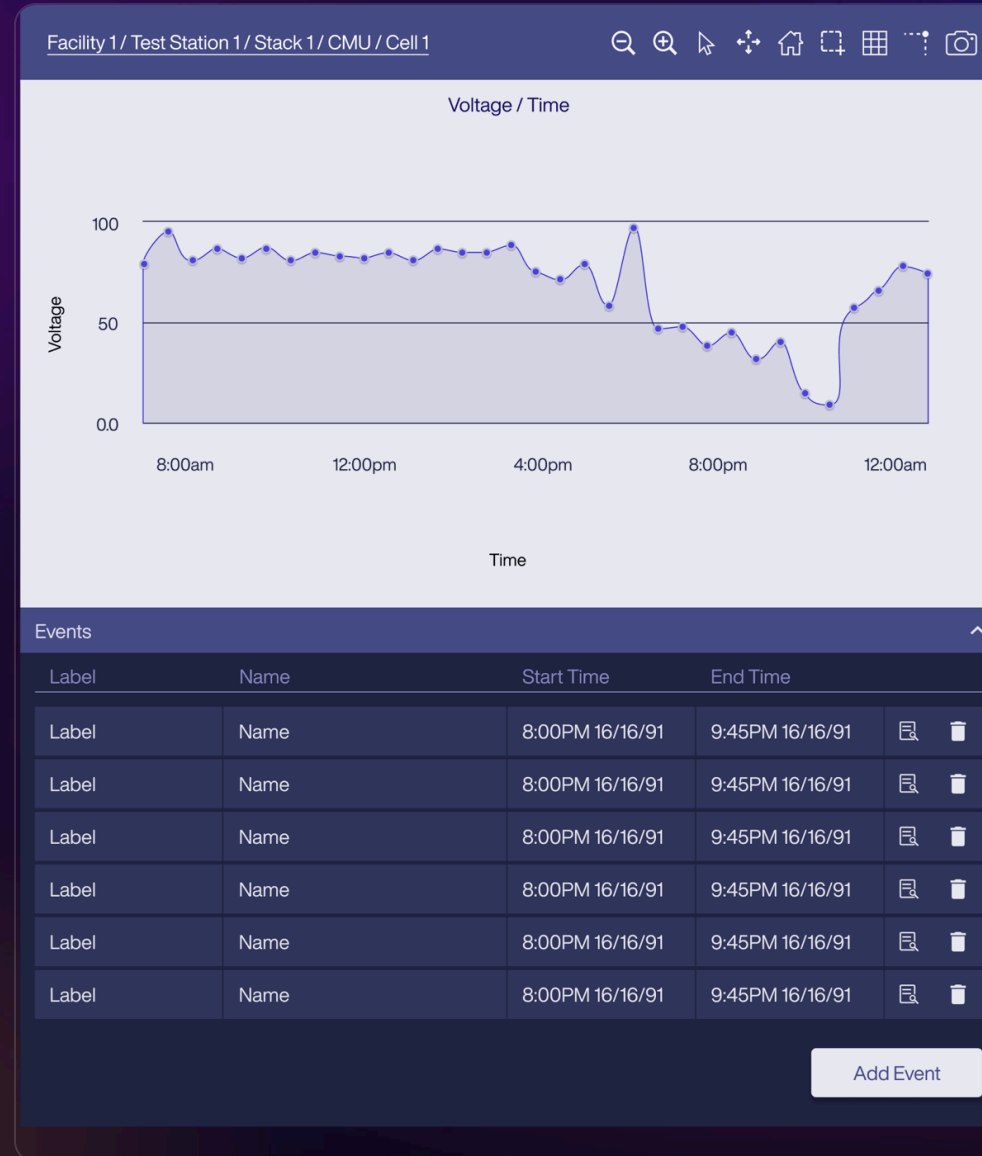
05

Work with the data in one place

Every graph shows its full hardware path at the top. You always know which cell you're looking at. If something odd shows up, you can select that range and save it as a tagged *Event*. The whole team can find it later. Events can also be automated: set a rule such as when a variable reaches a value, and the software flags it for you. Data uploaded from outside the probes sits in the same tables and filters as live probe data. Analysis sits next to the data, so you can work with it as soon as it arrives instead of exporting it first. Export is still there, but the built-in tools flag anomalies and let you tag events to points in time.

Adding a visualization reuses the hardware tree, then groups the data the way electrochemists think about it: electrochemical, frequency domain and time domain. Analysis tools such as correlation, regression and moving averages sit in the same app, each opening in its own tab.

Events on a graph, adding a visualization, and the analysis panel



Built as a system

Marc and I designed the UI system alongside the screens. It gave Pulsenics a set of reusable assets and made prototyping much easier.

Sections

Collapsible, resizable panels

Every area of the app (hardware, tags, live and saved experiments, data, visualizations) is a panel that can be opened, closed or resized. Users set the workspace up around what they're working on.

Navigation

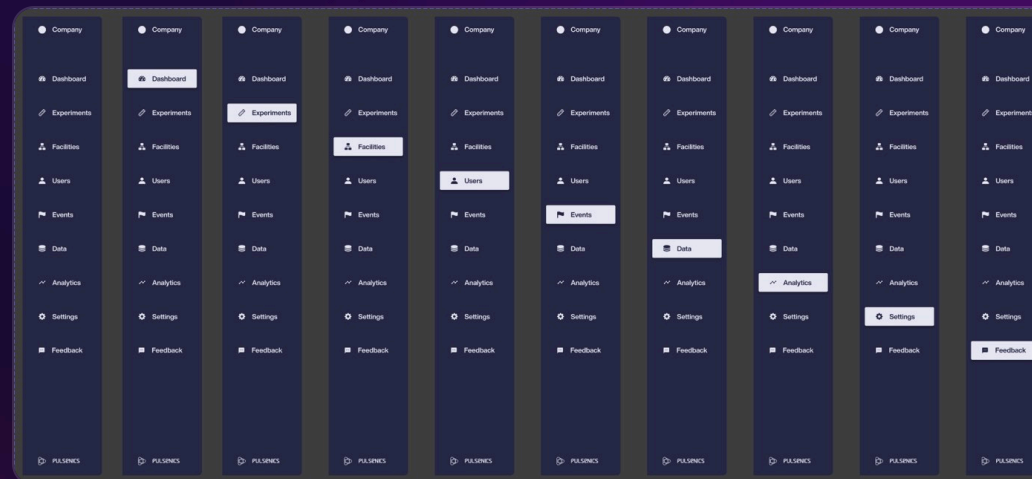
Nine sections, two widths

Dashboard, Experiments, Facilities, Users, Events, Data, Analytics, Settings and Feedback, each with a selected state, plus a slimmer rail with just the icons for more room.

Charts

One graph component

The same graph, toolbar and events tab are used everywhere data is plotted, in four states from no events to many.



Navigation, one state per section



Part of the component library: buttons, tabs and analysis chips



Space Cadet
#262845 · dashboard



Midnight Blue
#1B1463 · CTAs, focus,
header



Pacific Blue
#1DA1B2 · secondary brand



Turquoise
#4DDDC0 · tertiary brand



Cyber Yellow
#FED405 · tertiary brand



Magnolia
#E6E7F2 · light surfaces



✕ What I'd refine next

Contrast

Secondary text and column headers sit close to the panel colour. I'd raise them to meet WCAG AA, which matters for people staring at data all day.

Inputs on dark

The light input fields are the brightest thing on screen. A darker field with a clear focus ring would keep attention on the data.

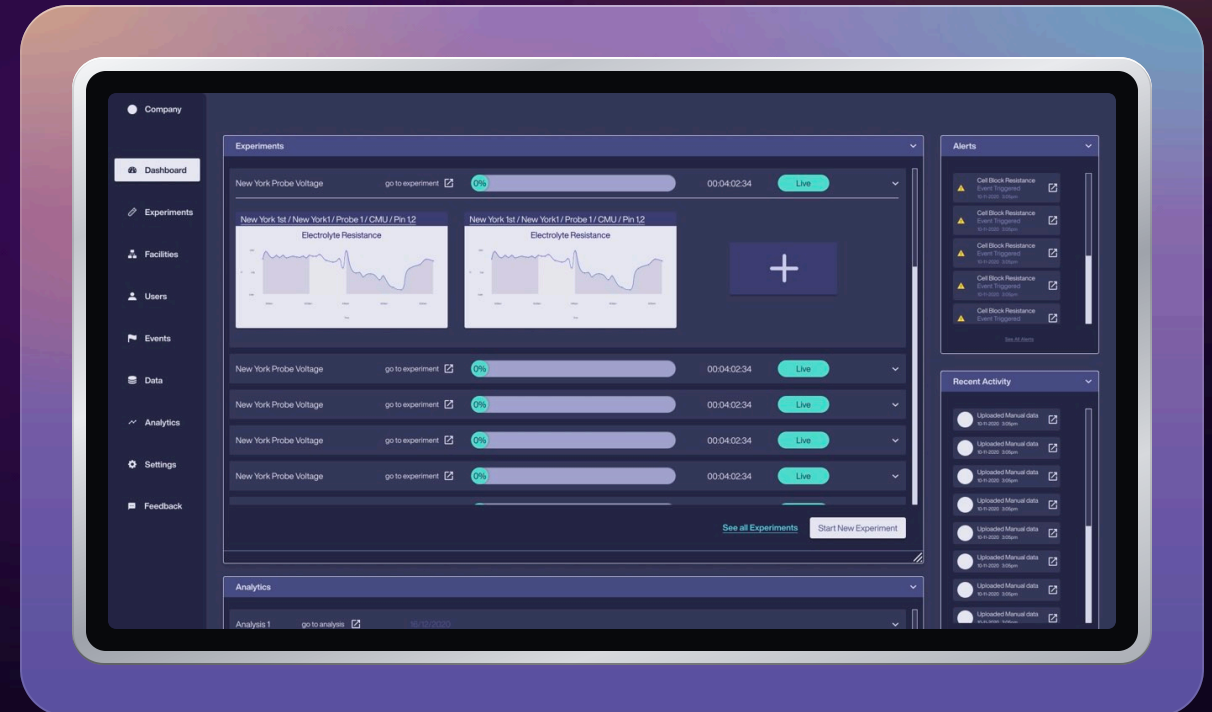
Hierarchy

Most panels carry the same visual weight. Giving live data more presence than setup controls would help scanning.

✕ Credits

Designed with Marc Varone

Pulsenics System Software was designed by two of us. We worked through every decision together, so the ideas on this page are shared.



[Read it on the site ↗](#)

[Next: Pulsenics Website](#)

Case study 03

Status

Live

Work

Web design and build, IA

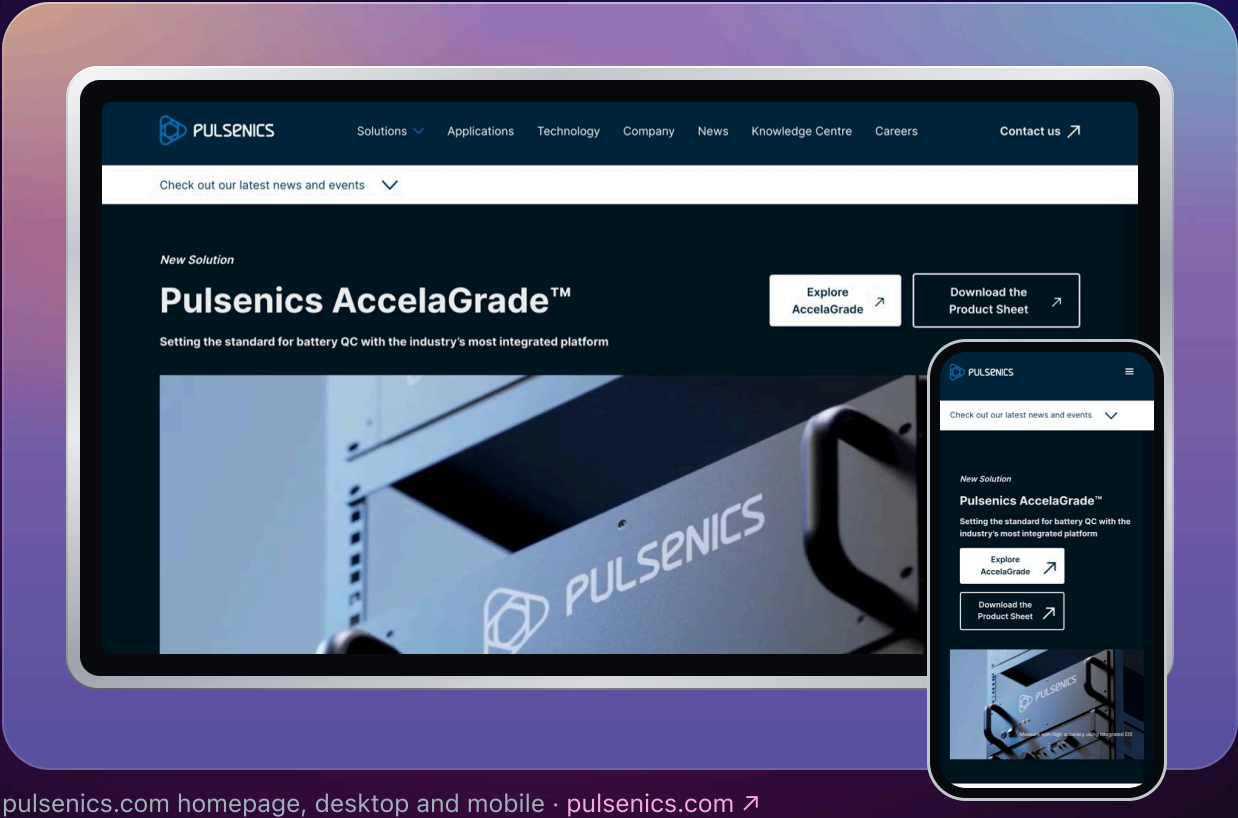
Platform

Webflow CMS

Pulsenics Website

I've worked on pulsenics.com since 2020. When Pulsenics wanted to add a lot of new pages, Marc Varone and I pushed for a workshop with their marketing team first. The Applications page came out of it, and it's now how visitors work out which product is right for them. I also built the Knowledge Centre, to improve the way Pulsenics publishes what it knows.

Client	Pulsenics, Toronto
My role	Design, build and upkeep since 2020; IA and homepage with Marc Varone
Platform	Webflow CMS
Editing	Set up so people without a technical background can publish their own content
Scope	Information architecture, homepage and product pages (with Marc Varone), Knowledge Centre, News
Status	Live, and I still maintain it



pulsenics.com homepage, desktop and mobile · pulsenics.com



✕ Overview

Problem

A lot of new pages to add

Before building them, we needed to know what a potential client should see when they land on the site.

What I did

A workshop, an Applications page and a Knowledge Centre

Marc and I ran a workshop with marketing that led to the Applications page. I also built the Knowledge Centre and set the site up on the CMS so staff without a technical background can publish.

Result

The main way clients find Pulsenics

The team updates it constantly with news and Knowledge Centre content, and it's the primary way potential clients find the company: Pulsenics' Google Analytics and marketing data show more people find them through Google search than through any other channel.

● The challenge

A technical product, for scientists and engineers

Pulsenics sells electrochemical impedance spectroscopy (EIS) hardware to battery, hydrogen and fuel cell teams. Its buyers are scientists and engineers. The site has to explain the products and keep up with a steady stream of news and studies.

I inherited the site from its original designer and kept its established look. Since then most of the pages have been redone or added as the company launched new products and needed new ways to explain its technology.

Information architecture · with Marc Varone

Help each visitor find the product they need

Pulsenics wanted to add a lot of new pages, and a new homepage hero. Rather than tack them on, Marc and I pushed for a workshop with their marketing team around one question: when a potential client lands on the site, what should they see, and how do we help them? The answer was that people didn't understand what each product was for. So we planned the site around two kinds of visitor.

From there we built two trees: one for the products and one for the industries they serve. Every industry splits into R&D and QC work, with room for case studies as Pulsenics publishes them.

Products

The Pulse Series

Continuous monitoring, many use cases

R&D

QC

AccelaGrade

Battery grading, one clear use case

Cell

Module

Software only

Industries

Battery

Applications: R&D, QC

Case studies

Hydrogen

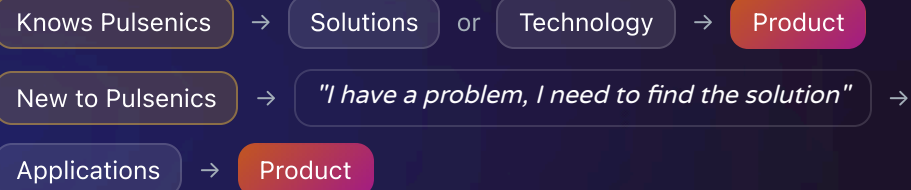
Applications: R&D, QC

Case studies

Specialty

Room for new kinds of client

Two ways in



From the workshop: the people who already know the products, and the people who only know their problem.

Mapping applications

Then we mapped every application to the person who would use the product, the person who would buy it, and the product that fits.

Application	End user	Buyer	Product
Factory cell and module QC	Test engineers	Head of QC, NPI managers	AccelaGrade
Remanufacturing QC	Process engineers	Head of factory operations	AccelaGrade
Cell and stack R&D	Research scientists	Head of R&D, Head of Technology	The Pulse Series

A few rows from the map. The full version covers battery, hydrogen and specialty work.

The Applications page

That map became the Applications page, which was our idea. It's where visitors decide which product is right for them. Each application points to one product, so nobody has to work out the product range on their own.

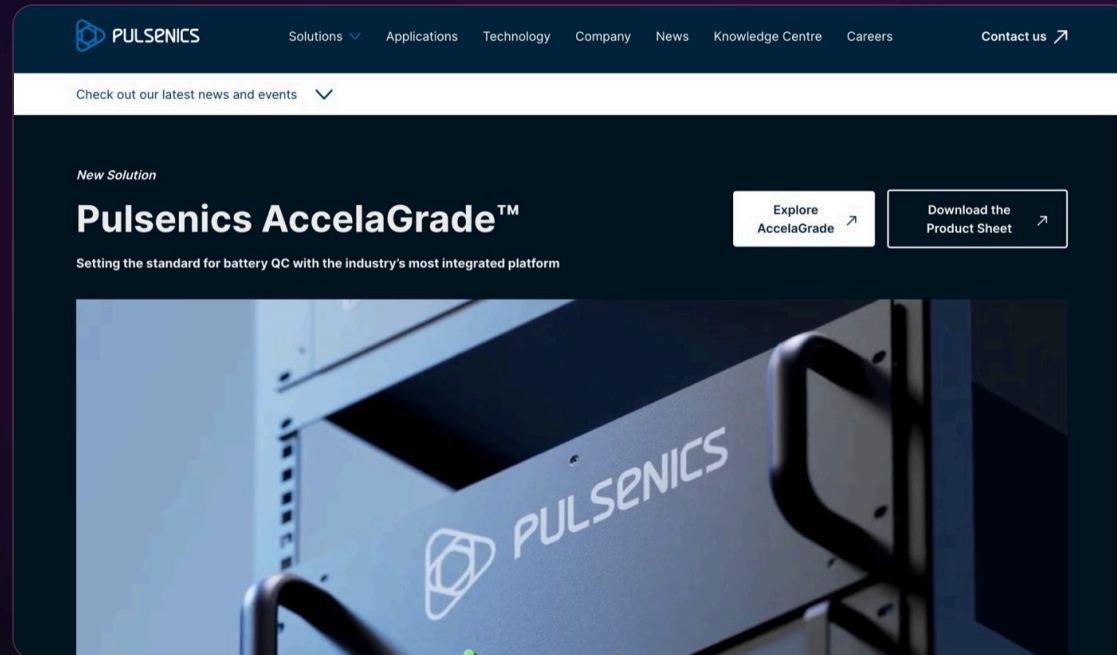
Battery	Module qualification · Second-life EV packs	AccelaGrade Module
Battery	Incoming cell QC · Remanufacturing for motive and micromobility	AccelaGrade Cell
Battery	Degradation benchmarking and diagnosis	The Pulse Series
Hydrogen	Stack failure analysis · Degradation benchmarking · Electrolyser stack defects	The Pulse Series

Homepage and product pages · with Marc Varone

01

Lead with the newest product

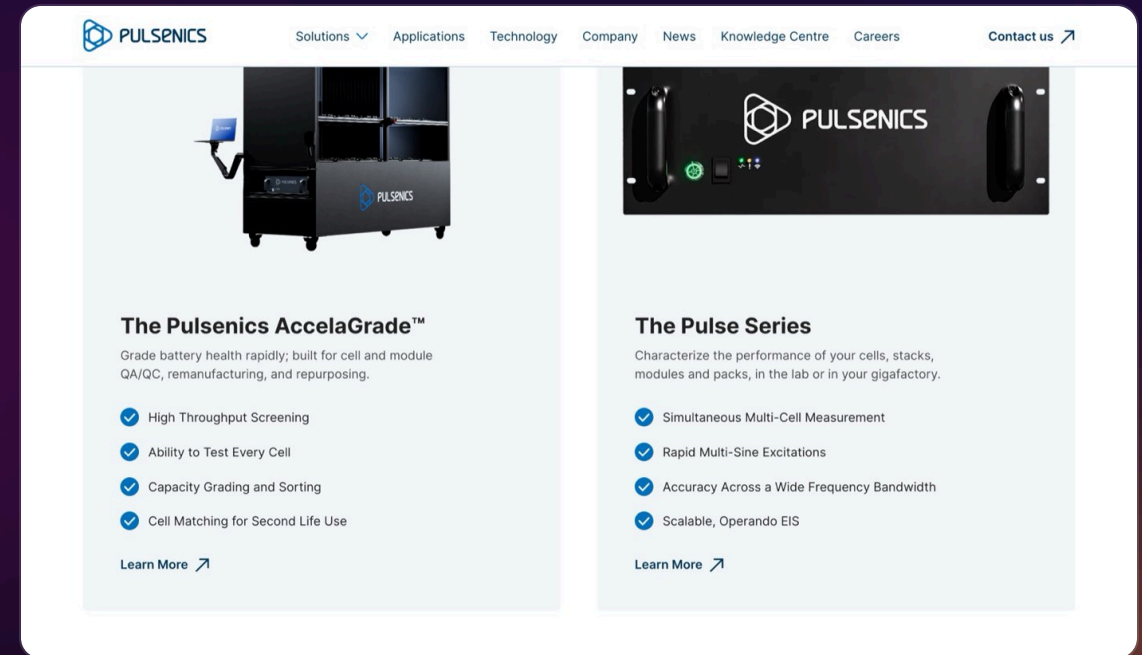
The homepage opens on AccelaGrade, the company's new battery grading system, with two clear actions: explore the product or download the spec sheet. It's their latest product, so it gets the top spot.



02

Two product lines, compared at a glance

AccelaGrade and the Pulse Series sit side by side in matching cards: a short line on what it's for, four capabilities each, and a single Learn More link. It's easy to see which one you need.



03

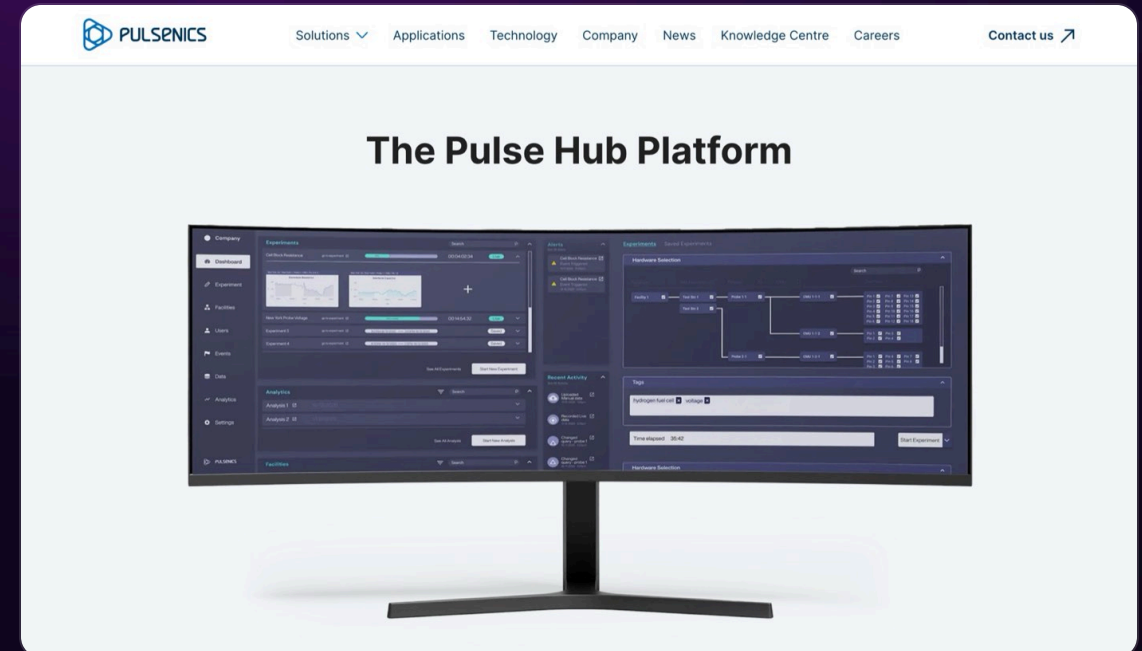
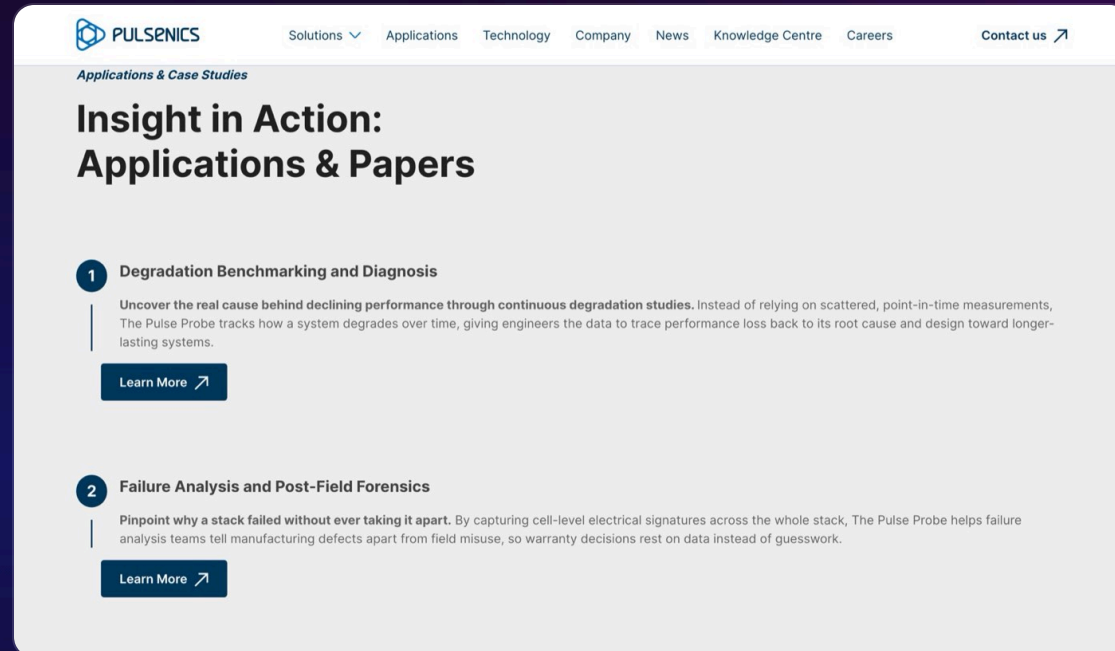
Product pages organised around real applications

The Pulse Series page explains the product through the problems it solves: degradation benchmarking, post-field failure analysis, catching stack defects before they ship. Each one links to a detailed Knowledge Centre article, so each product page leads into the detail behind it.

04

Give the software its own section

Every Pulse Series system ships with the Pulse Hub software, so the product page gives it a section of its own, shown on a monitor the way a lab would use it. The interface on that screen is the [System Software I designed with Marc Varone](#).



The Knowledge Centre

I built the Knowledge Centre to improve the way Pulsenics publishes what it knows about EIS. It reads like documentation: how the technology works, then how it's used in battery and hydrogen work. It's all built on the Webflow CMS, so anyone on the team can write and publish articles without touching the layout.

Left

Navigate by topic

Technology basics in order, then applications grouped by industry: battery and hydrogen.

Centre

Read in depth

Long articles with equations, figures and interactive charts inline. The Pulsenics team built the charts; I host them and embed them in the articles.

Right

Jump within a page

An automatic table of contents for each article, plus a link to talk to Pulsenics.


[Solutions](#)
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[Knowledge Centre](#)
[Careers](#)
[Contact us](#)

Pulsenics Knowledge Centre

Pulsenics Knowledge Centre

[Introduction](#)
[1. How does Electrochemical Impedance Spectroscopy work?](#)
[2. Signal Optimization](#)
[3. Interpreting EIS Data](#)
[4. Introduction to Multisine EIS](#)
[References](#)

Applications
 Battery
[Grade battery cells from motive and micromobility packs before redeployment](#)
[Grade new incoming cells before integrating into modules or packs](#)
[Grade incoming automotive packs before deployment into stationary storage applications](#)
[Qualify battery modules prior to pack integration](#)
 Hydrogen
[Diagnose stack failures and perform post-field forensics](#)
[Detect electrolyser stack defects and prevent quality escapes](#)

Welcome to the Pulsenics Knowledge Centre
 This resource serves as a comprehensive guide to understanding all facets of Electrochemical Impedance Spectroscopy (EIS), from the underlying science to real-world applications and measurement best practices.
 Whether you're new to impedance analysis or an experienced electrochemist looking to deepen your expertise, this Knowledge Centre is designed to meet you where you are. We've organized content to take you from foundational principles, how and why impedance measurements work, through to advanced topics like equivalent circuit modelling, data interpretation, and troubleshooting common artefacts.
 EIS is one of the most powerful and information-rich techniques in electrochemistry, capable of probing processes across many orders of magnitude in frequency. But that power comes with complexity. Our goal here is to make that complexity approachable: clear explanations, practical guidance, and worked examples grounded in the kinds of problems our users actually encounter. Inside, you'll find:
 1. **Science & Theory:** The physics and mathematics of impedance, from basic circuit analogies to the Kramers-Kronig relations
 2. **Instrumentation & Setup:** How to configure your system for reliable, low-noise measurements
 3. **Applications:** How EIS can be used across a range of fields, including batteries, fuel cells, and electrolysers
 4. **Data Analysis:** Fitting strategies, model selection, and common pitfalls
 5. **Best Practices:** Practical tips for getting the most out of every measurement
 This is a living resource, updated as the science evolves and as we learn from our community. If you have questions or topics you'd like to see covered, we'd love to hear from you.

[Speak with Pulsenics](#)

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Knowledge Centre home


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Pulsenics Knowledge Centre

3. Interpreting EIS Data

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3.0.1 The Nyquist Plot
 As previously mentioned in Module 1, the Nyquist plot is one of the most common ways to quickly visualize and interpret raw EIS data. Based on the type of electrochemical system and its state, the shape of the Nyquist plot may differ. As temperature, state of charge, and health change, we can observe the differences in the Nyquist data. The interactive figure below shows the effect of state of charge on the Nyquist plot's signature.

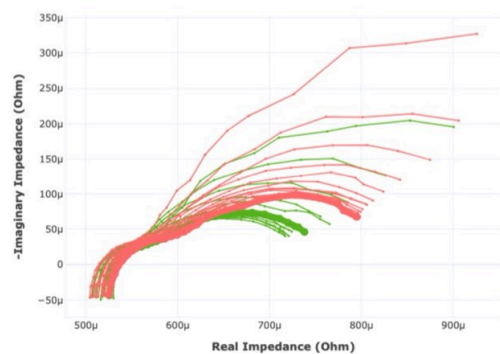


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[3.0.1 The Nyquist Plot](#)
[3.0.2 Distribution of Relaxation Times](#)
[3.0.3 Example Equivalent Circuit Model Process](#)
[Want to learn more? Speak with us today!](#)

An article with an interactive chart

 News

Pulsenics publishes partnerships, conference talks, product launches and studies almost every month. The News page has a featured story on top and a filterable grid underneath: Announcement, Conference, Technology, Company, Collaboration, and Studies & Papers.

It runs on the CMS like the Knowledge Centre. Someone on the team adds a story, and it shows up on the News page and in the latest news banner at the top of the homepage. Nobody needs me to keep the site current.

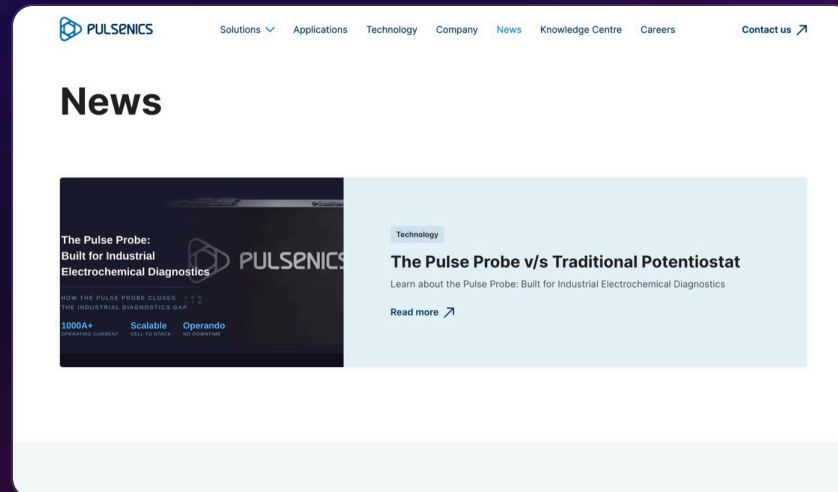
 See it live

Live, and updated all the time

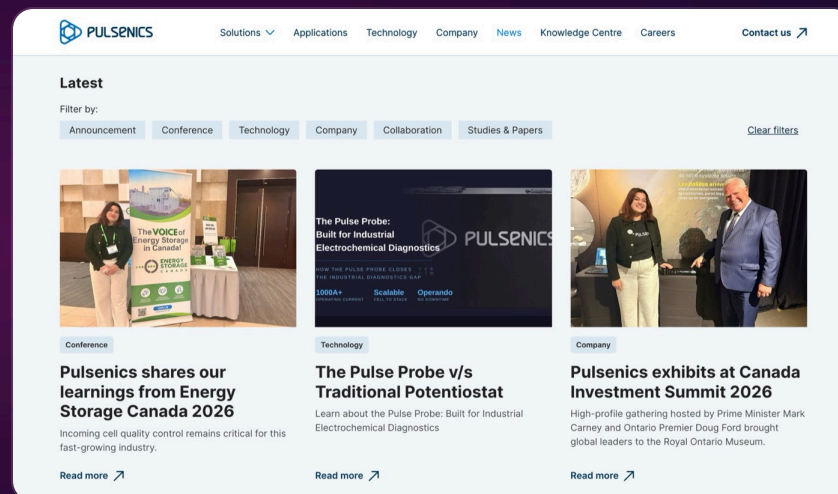
The team keeps it updated with news and Knowledge Centre content, and I keep building on it as Pulsenics launches products and publishes research.

Looking back

I'd push for the new structure sooner. Once marketing joined the workshop and we mapped who needs what, the changes came quickly; I'd make that case on day one.

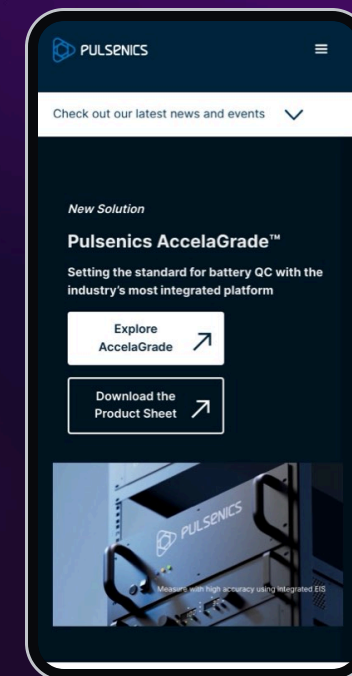


Featured story



Category filters and the article grid

 On a phone



Homepage. Actions stack full width.



Articles. Embedded charts still work on a phone.

Every page is responsive, including the articles and the charts inside them.

Earlier work

Status
LiveWork
Website design and buildPlatform
Webflow

Gabion Developments

A marketing site for a Toronto design-build firm. It puts their finished homes up front and makes booking a call the obvious next step.

Client Gabion Developments, Toronto

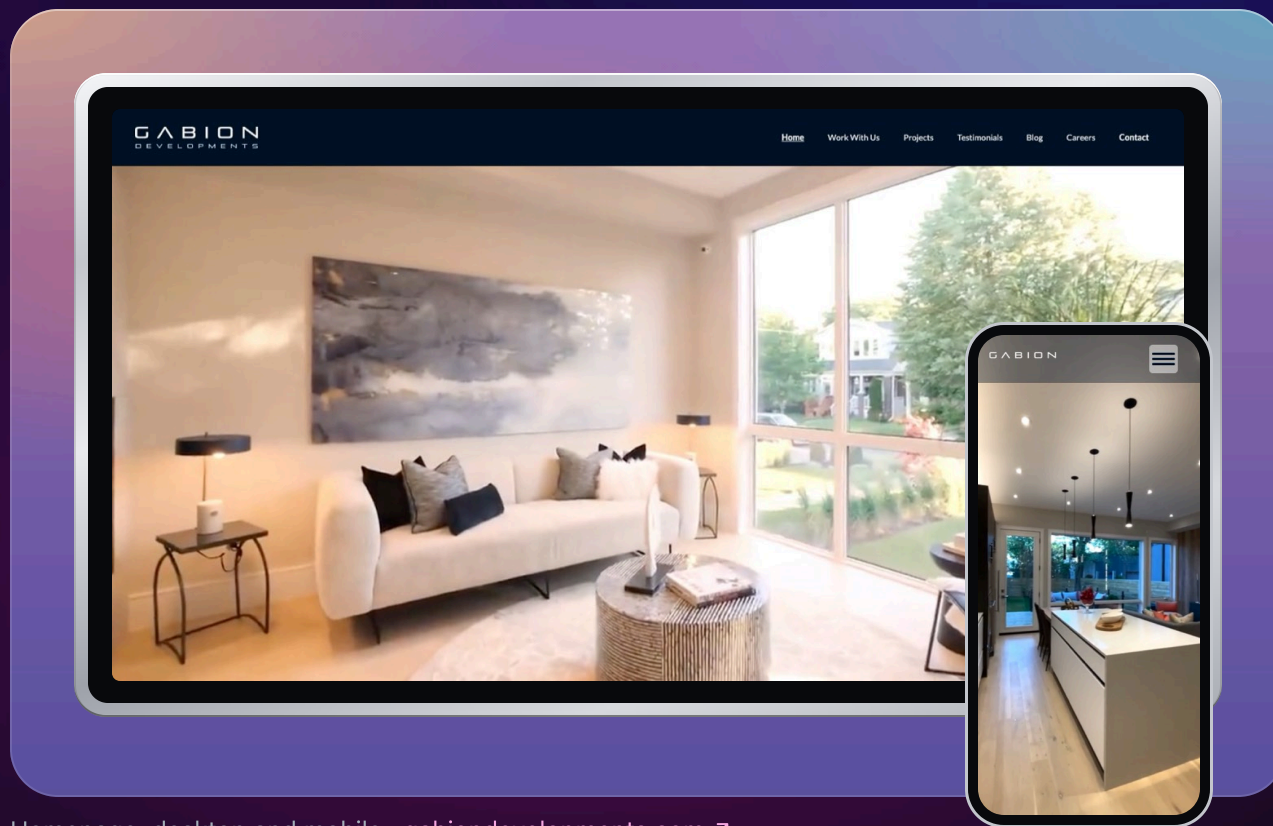
My role Sole designer and site builder

Platform Webflow

Timeline 2 months, concept to launch

Launched 2022

Scope IA, UX/UI, visual design, responsive build



Homepage, desktop and mobile · gabiondevelopments.com

A site in the firm's own style that markets its homes and gives people a professional way to reach them. I designed and built it alone: structure, visual design and a responsive Webflow build, in two months, with the owners and their marketing lead. It launched in 2022. The owner told me it was exactly what he'd envisioned, with the look and feel he wanted to attract the kind of client he was after.

● The brief

Gabion needed a site to market its home developments, and to give prospective clients a professional way to find them.

They wanted a site that spoke to their own design style, one that let their work speak first. That's what I led with: a video of a finished home on the homepage, and minimal styling so the photos and video they provided carry the page.

● What the site had to do

Sell with finished work

People judge a builder by what it has already built, so the site had to put real homes in front of visitors straight away.

Make the process feel manageable

Building or renovating a home is a big, unfamiliar decision. The site needed to show what happens, and in what order, before anyone picks up the phone.

Turn interest into a conversation

One clear next step, repeated across the site: book a call.

How we worked

Two months, with feedback the whole way through

We started with a kickoff with the owners. After that I worked with their marketing lead on the rest of the site, with the owners checking in along the way. They gave feedback continuously and supplied the content: the photos and videos of their homes.

1 Structure

Mapped the seven pages and the path from a first visit to Book a Call.

2 Design

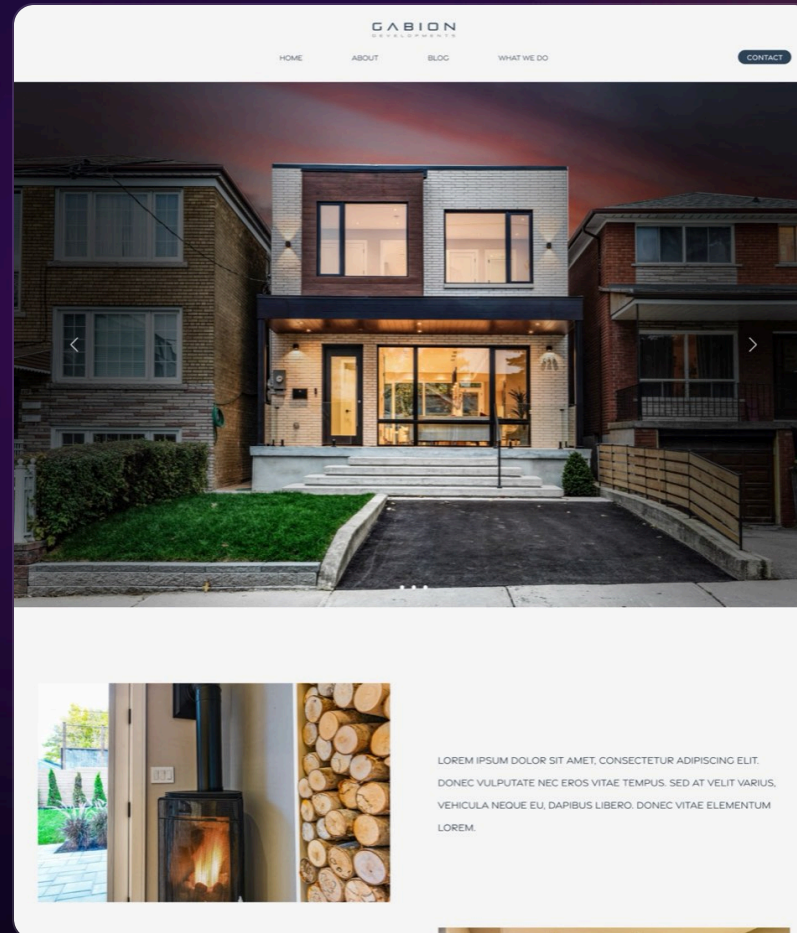
Set the typography, grid and image treatments so the photography leads.

3 Build

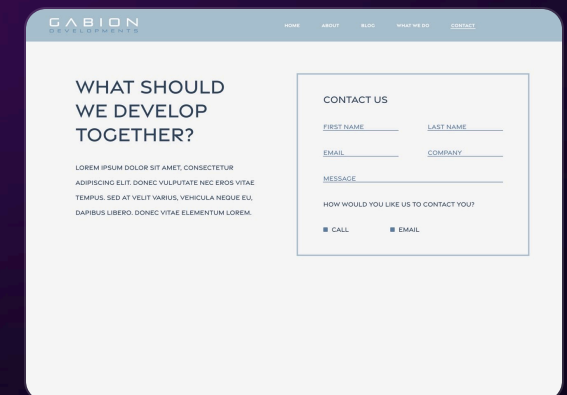
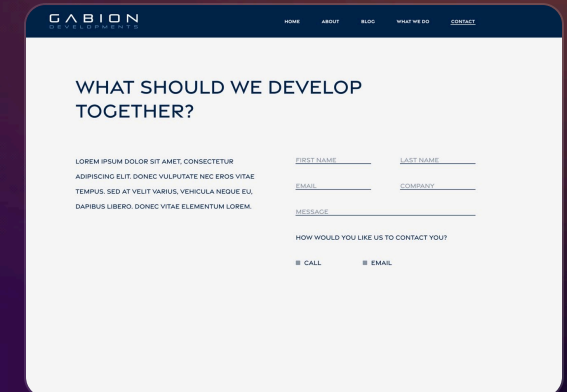
Built responsive pages in Webflow from reusable sections, so the firm's team can edit copy and add projects without a developer.

4 Launch

Checked every page across screen sizes, then went live.



Early concept. Home v1 opened on a photo carousel. The live site moved to a video of a finished home.



Two directions for the contact page



Information architecture

A flat structure with seven pages

The only nested level is the individual project pages, so nothing sits more than two clicks from the homepage.

Home	Video hero, welcome, philosophy, Book a Call banner
Work With Us	The journey, in four stages
Projects	Selected work and upcoming projects
Testimonials	Homeowner quotes; pick a name to read each one
Blog	Image gallery linking to Instagram
Careers	Join Our Team, with a link to current openings
Contact	Enquiry form with a "How did you hear about us?" field

Projects: Selected work

Finished projects, each with its own page.

Cedarvale

Upper Beaches

Bayview

Projects: Upcoming

Projects in the pipeline, shown as architectural drawings.

Laneway

Trinity Bellwoods

Beaches

Oakwood

East York

Port Credit

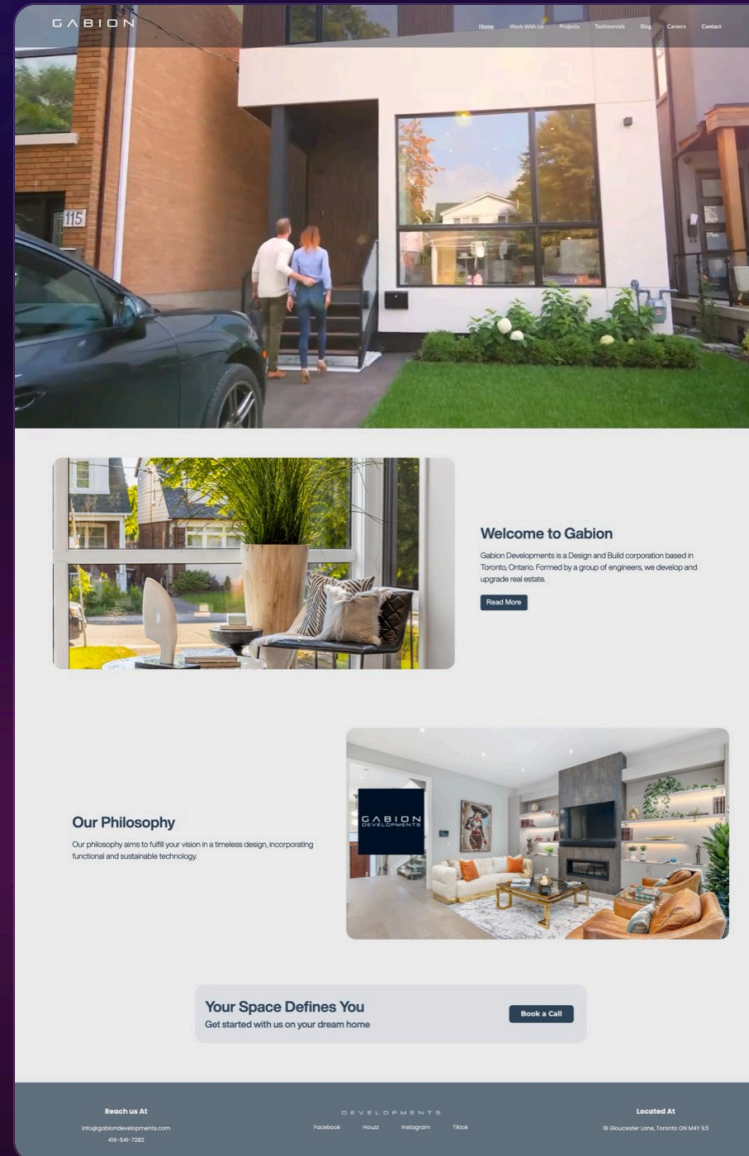
Port Union

Six design decisions

01

Open with a finished home

The homepage opens on a video of a completed project, 111 Malvern Ave, that fills the whole screen. There's very little copy. Before any copy about the company, you see one of their homes. The welcome and philosophy sections come after that.

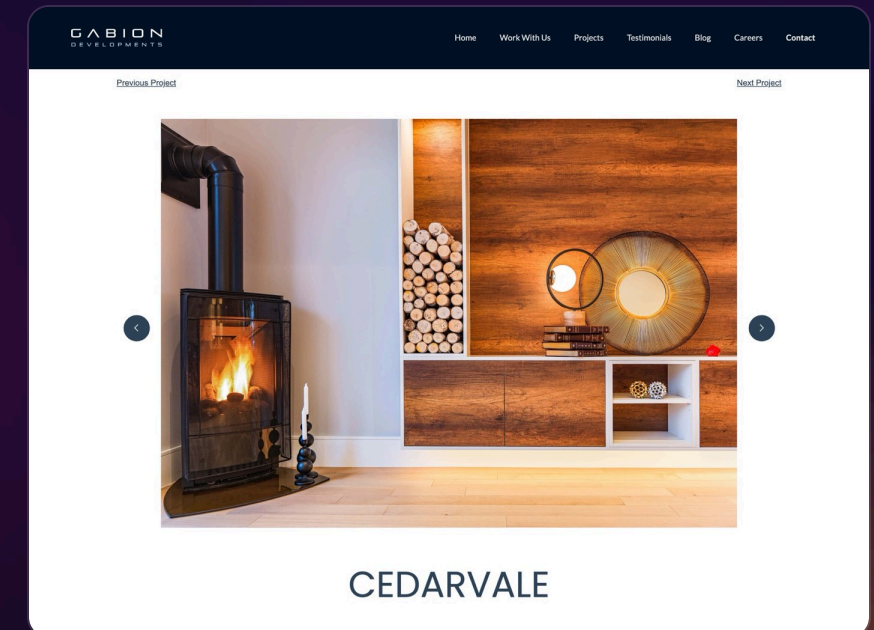


Homepage, full length

02

Let the photography carry the design

I kept the interface simple so the photos could do the work. Every page is built around large images, with plenty of white space and very little else competing for attention. Each project page leads with a photo carousel, keeps the description short, and links to the previous and next project so visitors can keep browsing.

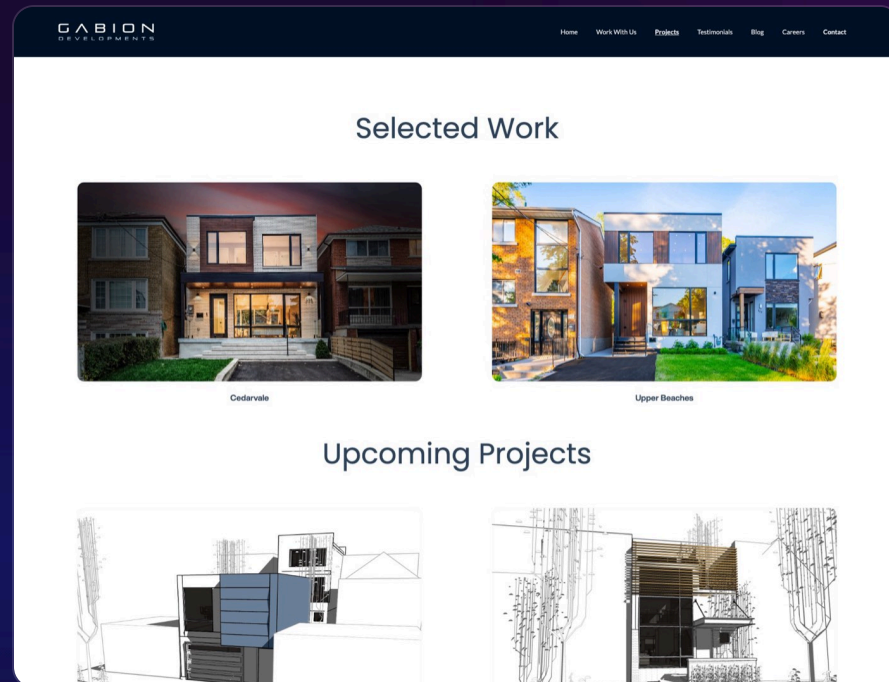


Project page: Cedarvale

03

Separate what is built from what is coming

The Projects page splits Selected Work from Upcoming Projects. Finished homes are shown in photographs and each gets its own page. Upcoming ones are shown as architectural drawings, so the difference is clear at a glance. Visitors can see what's finished, and that there's more on the way.



Projects: photos for finished work, drawings for what is coming

04

Show the process as four stages

Work With Us walks through Exploring, Imagining, Creating and Finalizing. Each stage explains what Gabion does at that point, such as visualizing design options and submitting the permit while Imagining, or monitoring construction while Creating. The images move from a pencil sketch to a rendering, a framing photo and the finished house, so the page shows the project taking shape.

It shows a new client what happens after they get in touch.

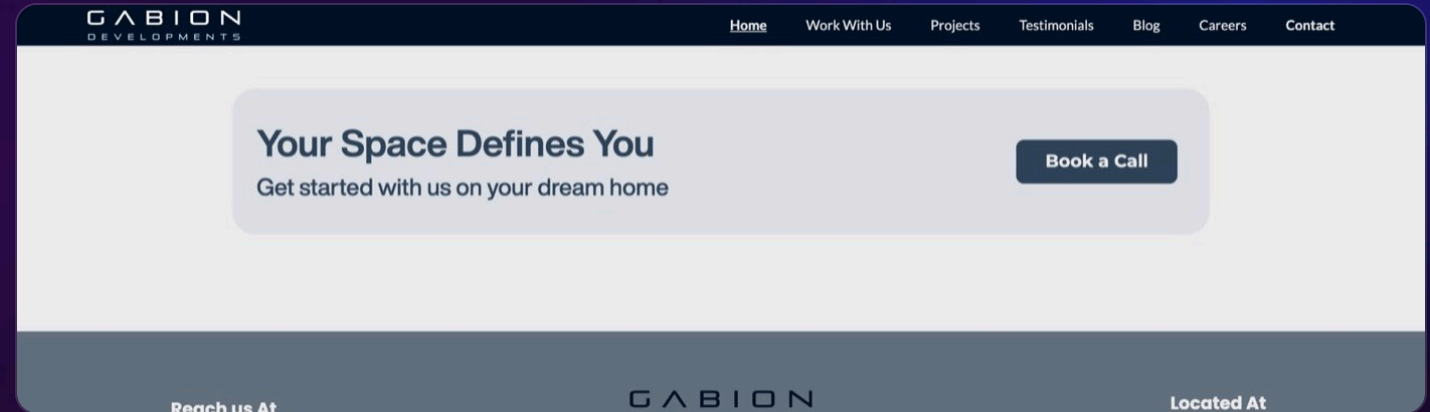


Work With Us: the journey, from sketch to site

05

One call to action, repeated

Book a Call is the main way to start. A banner reading “Your Space Defines You” closes almost every page, from the homepage to each project and the contact page, so there's always a way to get in touch close by. The footer below it repeats the email, phone number and address.

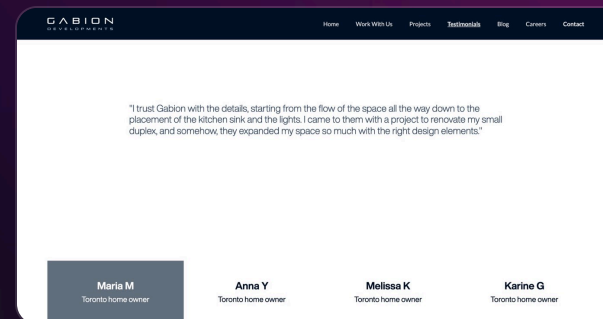


The Book a Call banner and footer, repeated across the site

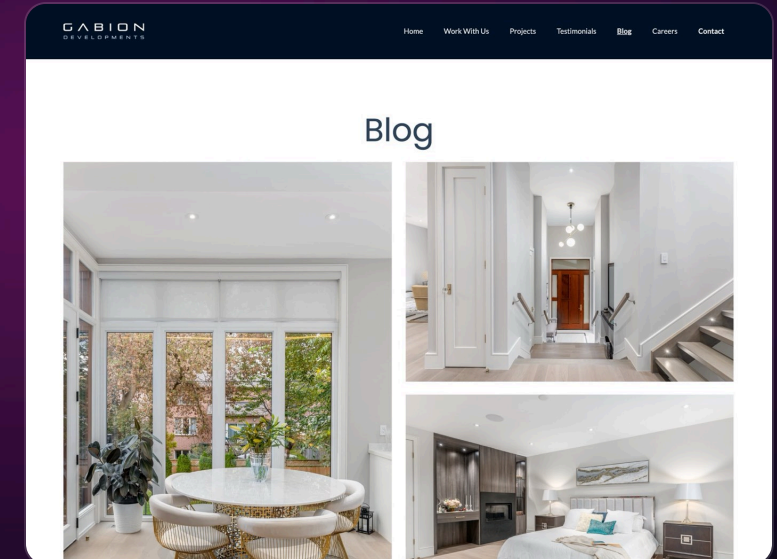
06

Proof from clients, and a blog that runs on Instagram

Testimonials gives homeowners' own words their own page: one quote at a time on desktop, chosen from a row of names, and the full list of quotes stacked on mobile. The Blog is a gallery of image tiles that link out to the firm's Instagram posts, so the team only has to post once and the site stays up to date.



Testimonials



Blog

● The same site on a phone

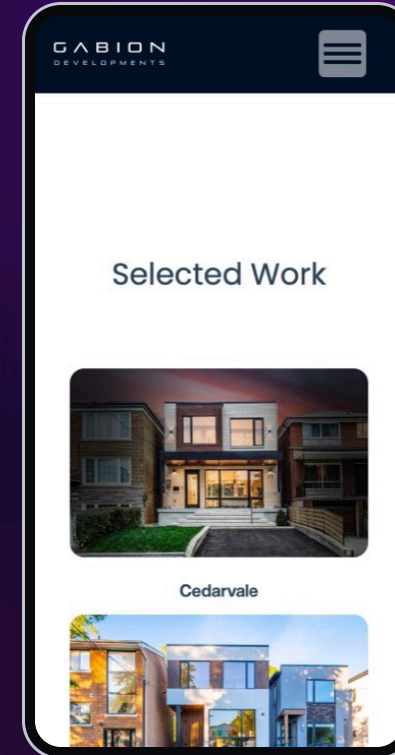
The navigation collapses behind a menu button, paired layouts stack image over text, and grids drop to one column. Components that rely on a wide screen change form: the testimonial selector becomes a simple list of every quote.



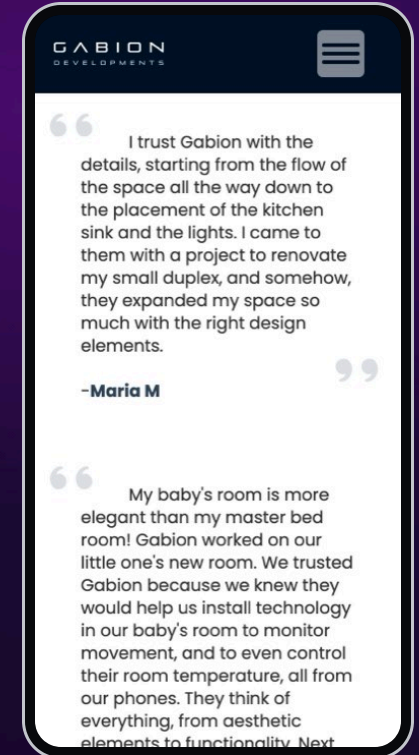
Home. The menu folds into one button.



Work With Us. Each stage stacks its image above the text.



Projects. Two columns drop to one.



Testimonials. The name selector becomes a list of quotes.

✕ The result

Launched in 2022, and the client loved it

The owner told me it was exactly what he'd envisioned: it had the look and feel he wanted, to attract the kind of client he was after. I handed it over after launch and don't maintain it now.

[Visit gabiondevelopments.com](https://gabiondevelopments.com)

[Contact](#)



● Contact

Hiring for a product or UX role?

I'm open to roles on data-heavy B2B products and internal tools.

ahmad.gh.ibrahim@gmail.com[LinkedIn ↗](#)[Request resume ↗](#)[ahmad-ibrahim.com ↗](http://ahmad-ibrahim.com)